

# STAGE 10: PDF GENERATION & VALIDATION ENGINE VERIFICATION REPORT

System Version: 1.0.0 | Verification Date: September 29, 2026 | Author: ARM Quality Assurance & PDF Conversion Engine Team | Overall Status: **PASS**  
(READY FOR STAGE 11)

## 1. EXECUTIVE SUMMARY & STAGE OBJECTIVE

Stage 10 implements the server-side PDF generation and structural validation engine for ARM (AI Academic Report Assistant). The objective of Stage 10 is to convert the validated **Stage 9 DOCX report** into a publication-grade PDF while preserving document hierarchy, typography, margins, tables, images, bullet and numbered lists, running headers, and footers. The pipeline strictly operates server-side in an isolated temporary workspace with timeout enforcement, PyMuPDF parser reopen testing, content consistency checking against DOCX, secret leakage scanning, and visual page rendering.

## 2. KEY ARCHITECTURAL GUARANTEES & VERIFIED RESULTS

| Architectural Pillar                        | Engine Implementation & Quality Gate                                                                                                                                                            | Verification Status               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Server-Side Conversion & Isolation          | Conversion executes purely server-side in isolated temporary workspaces with zero browser or user Word dependency. Workspace is guaranteed to be deleted after completion or failure.           | VERIFIED & SECURED                |
| Converter Abstraction & Fallback            | PdfConverterService prioritizes LibreOffice headless when present, seamlessly falling back to PurePythonDocxPdfConverter for zero-dependency standalone execution in all environments.          | VERIFIED (Dual Converter)         |
| Converter Availability & Timeout Protection | Verifies converter availability before execution (HTTP 503 if unavailable). Conversion process enforces strict bounded timeouts (HTTP 504 on timeout), terminating child processes cleanly.     | VERIFIED & PROTECTED              |
| PyMuPDF Reopen & Structural Validation      | Opens output PDF with PyMuPDF (fitz) and pypdf. Confirms valid magic header (%PDF-), page count > 0, full text extractability across all pages, and structural integrity.                       | VERIFIED (PyMuPDF Reopened)       |
| DOCX & Template Immutability                | Source Stage 9 DOCX SHA-256 and original college template SHA-256 are verified unchanged before and after PDF conversion. Zero mutation of source assets.                                       | VERIFIED (Hashes Intact)          |
| Content Consistency & Anti-Bleed            | Ensures project title, major headings, representative paragraphs, tables, and references from Stage 9 DOCX exist in PDF. Distinct projects and templates produce isolated PDFs with zero bleed. | VERIFIED & CONSISTENT             |
| Visual Page Rendering Validation            | PyMuPDF renders representative pages (first, content, table, final) to pixmaps at 96 DPI. Inspects pixel samples to verify no blank pages, abnormal dimensions, or visual clipping.             | VERIFIED (Visual PASS)            |
| Multi-Tenant IDOR & Secret Leakage          | Multi-tenant project ownership is enforced (User B receives HTTP 403 on User A's PDF). Content is scanned for forbidden secrets (Gemini keys, Supabase tokens, DB credentials).                 | VERIFIED (No Leakage / IDOR Safe) |

## 3. AUTOMATED TEST SUITE EXECUTION RESULTS

| Test Suite / Stage                     | Scope & Subsystems Covered                                                    | Tests     | Result      |
|----------------------------------------|-------------------------------------------------------------------------------|-----------|-------------|
| Stage 1 — Database Foundation          | PostgreSQL schema, RLS policies, profiles, generation_jobs                    | 7 / 7     | 100% PASS   |
| Stage 2 — Template Ingestion           | DOCX/PDF upload, MIME validation, private storage                             | 10 / 10   | 100% PASS   |
| Stage 3 — Template Intelligence        | Deterministic OpenXML AST parsing, TemplateSchema normalization               | 9 / 9     | 100% PASS   |
| Stage 4 — Template Review + Lock       | Patching template schema, hierarchy validation, lock immutability             | 10 / 10   | 100% PASS   |
| Stage 5 — Project Evidence Locker      | Fact storage, project members, file evidence, readiness indicator             | 8 / 8     | 100% PASS   |
| Stage 6 — AI Report Planner            | ReportPlan generation, section breakdown, Gemini integration                  | 7 / 7     | 100% PASS   |
| Stage 7 — Report Content Generator     | Section-by-section Gemini generation, grounding status, versioning            | 10 / 10   | 100% PASS   |
| Stage 8 — Citation Verifier            | Claim extraction, evidence mapping, anti-fabrication, quality gates           | 12 / 12   | 100% PASS   |
| Baseline Services & Endpoints          | Health check, document assembler baseline, template parser                    | 12 / 12   | 100% PASS   |
| Stage 9 — Deterministic DOCX Engine    | DocumentAssembler, DocumentValidator, SHA-256 immutability, IDOR              | 11 / 11   | 100% PASS   |
| Stage 10 — PDF Generation & Validation | Server-side conversion, PyMuPDF reopen, visual validation, IDOR, immutability | 16 / 16   | 100% PASS   |
| TOTAL INTEGRATED TEST SUITE            | Zero regressions across all stages 1 through 10                               | 112 / 112 | 100% PASSED |

4. STAGE 10 DEDICATED VERIFICATION SUITE BREAKDOWN (16/16 PASSED)

| Test Identifier                                   | Tested Requirement                  | Verified Behavior                                                                                                                  | Result |
|---------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|
| test_mandatory_pdf_conversion_and_reopen_test     | Mandatory PDF Test (Sec 51.1, 51.7) | Converts Stage 9 DOCX to PDF; reopens with PyMuPDF, verifies page count > 0, extracts text, finds title, headings, and references. | PASS   |
| test_corrupted_docx_rejected                      | Corrupted Input (Sec 51.2)          | Non-ZIP / corrupt DOCX rejected prior to conversion with HTTP 400 Bad Request.                                                     | PASS   |
| test_missing_docx_safe_failure                    | Missing DOCX (Sec 51.3)             | Request for uncompiled version returns HTTP 400 Bad Request safely.                                                                | PASS   |
| test_converter_unavailable_handling               | Converter Unavailable (Sec 51.4)    | Simulated unavailable converter returns HTTP 503 Service Unavailable with clear infrastructure message.                            | PASS   |
| test_converter_timeout_handling                   | Converter Timeout (Sec 51.5)        | Exceeding timeout limit terminates safely, cleans temporary workspace, and returns HTTP 504 Gateway Timeout.                       | PASS   |
| test_invalid_pdf_output_rejected                  | Invalid PDF Output (Sec 51.6)       | Corrupt/zero-byte converter output rejected by structural validator with HTTP 422.                                                 | PASS   |
| test_content_consistency_docx_to_pdf              | Content Consistency (Sec 51.8)      | Project title, major headings, representative content, and references verified present in PDF.                                     | PASS   |
| test_different_projects_distinct_pdf_content      | Project Isolation (Sec 51.9)        | Attendance project PDF vs Agriculture project PDF produce mutually exclusive content with zero bleed.                              | PASS   |
| test_different_templates_distinct_pdf_layout      | Template Fidelity (Sec 51.10)       | Template A (Times New Roman) vs Template B (Arial) produce distinct PDF binaries and layouts.                                      | PASS   |
| test_source_docx_and_master_template_immutability | Immutability (Sec 51.11, 51.12)     | Source Stage 9 DOCX and master template SHA-256 verified identical before and after PDF conversion.                                | PASS   |
| test_multi_tenant_idor_protection                 | IDOR Protection (Sec 51.13)         | User B denied compile and download access to User A's PDF (HTTP 403 Forbidden).                                                    | PASS   |
| test_path_traversal_and_filename_sanitization     | Path Traversal (Sec 51.14)          | Directory traversal sequences ('../..') sanitized safely in download filenames.                                                    | PASS   |
| test_secret_leakage_detection                     | Secret Leakage Scan (Sec 51.15)     | PDF content scanned for Gemini keys, Supabase service keys, and database passwords (0 detected).                                   | PASS   |
| test_repeated_generation_idempotency              | Idempotency (Sec 51.16)             | Repeated PDF compilation for same version reuses canonical path without uncontrolled duplicate files.                              | PASS   |
| test_visual_rendering_validation                  | Visual Validation (Sec 52)          | PyMuPDF renders representative pages at 96 DPI, verifying no blank pages, clipping, or abnormal dimensions.                        | PASS   |
| test_api_compile_and_download_flow                | Full API Flow (Sec 51.18)           | POST /compile-pdf compiles PDF; GET /download-pdf streams PDF binary with Content-Disposition headers.                             | PASS   |

5. FRONTEND VERIFICATION & PRODUCTION BUILD

- Frontend Verification Status:
- **TypeScript Typecheck:** tsc -p apps/web/tsconfig.json --noEmit completed with **0 errors**.
  - **ESLint Verification:** npm run lint completed with **0 errors and 0 warnings**.
  - **Next.js Production Build:** npm run build successfully compiled all **26/26 routes** with static optimization.
  - **UI Component Integration:** Enhanced DocumentCompiler (apps/web/src/components/arm/document-compiler.tsx) supporting both Stage 9 DOCX and Stage 10 PDF generation, live progress steppers, PyMuPDF validation badges, and direct downloads.

6. STAGE BOUNDARY & SIGN-OFF

**STAGE 10 COMPLETION DECLARATION**

Server-Side PDF Generation & Validation Engine is fully implemented, verified, and locked.

- **Stage 10 Status:** **PASS**
- **Total Tests Passed:** 112 / 112 (0 Regressions across Stages 1–10)
- **Source DOCX & Template Immutability:** VERIFIED (SHA-256 Hashes Intact)
- **PyMuPDF Reopen & Text Extraction:** VERIFIED (Page count > 0, full text extractable)
- **Visual Page Rendering Validation:** VERIFIED (PASS — 0 blank pages, 0 clipping)
- **Stage Boundary Enforcement:** Report workspace redesign, editing workflow, and weekly report automation have NOT been implemented (strictly deferred to future stages).
- **READY FOR STAGE 11:** **YES**